

SHIH-WEN LIU (CASPER)

RESEARCH INTERESTS  casperliu1118@gmail.com |  [casperliuliuliu.github.io](https://github.com/casperliuliuliu)

My core interests are **cooperative and scalable robotic intelligence**. I particularly focus on designing learning mechanisms that enable multiple robots to coordinate and collaborate efficiently, especially in multi-agent robotic manipulation and control. Previously, I worked on parameter-efficient fine-tuning for multi-task learning, and histopathology report generation.

EDUCATION

- **National Cheng Kung University** Sep. 2024 – Present
Master of Science in Artificial Intelligence
◦ Overall GPA: 4.0/4.0, Advisor: Prof. Wei-Ta Chu Tainan, Taiwan
- **National Sun Yat-sen University** Sep. 2020 – Jun. 2024
Bachelor of Science in Computer Science Engineering
◦ Last 60 GPA: 3.9/4.0 Kaohsiung, Taiwan
◦ Academic Excellence Award (2020)

PUBLICATIONS

- [C.2] **Shih-Wen Liu, Yen-Chang Chen, Wei-Ta Chu, Fu-En Yang, Yu-Chiang Frank Wang "Frequency Switching Mechanism for Parameter-Efficient Multi-Task Learning."** In *Proceedings of the IEEE/CVF International Conference on Computer Vision (CVPR) 2026*.
- [C.1] **Shih-Wen Liu, Hsuan-Yu Fan, Wei-Ta Chu, Fu-En Yang, Yu-Chiang Frank Wang "Histopathology Image Report Generation by Vision Language Models with Multimodal In-Context Learning."** In *the Conference of Medical Imaging with Deep Learning (MIDL) 2025*.

PROJECTS

- **Split Conv: Efficient Convolution Layer Design** Aug. 2024
Tools: Pytorch [Project]
◦ Designed memory-efficient convolution and pooling layers for large-resolution training
◦ Optimized implementations to reduce VRAM usage by 50 times
- **Jetson Nano Real-Time AR Teaching System** Jun. 2024
Tools: Python, OpenCV, Jetson [Demo Video]
◦ Developed AR overlay pipeline on Jetson Nano for live video processing
◦ Achieved 30 FPS on 720p streams with real-time object detection
- **Desiary - A Novel Diary-Dating App** May 2024
Tools: Flask, React Native, Stable Diffusion, LoRA, MongoDB, Docker [Demo Video]
◦ Created diary-to-image generation using Stable Diffusion with LoRA fine-tuning
◦ Built full-stack mobile app with user authentication and image gallery
◦ Deployed backend in Docker with CI/CD pipelines for reliability
- **Newsbie - Your Little News Reporter** May 2024
Tools: Python, Flask, React Native, MongoDB, Docker [Demo Video]
◦ Implemented web-scraping modules to aggregate news from 10+ sources
◦ Integrated LLM API for concise, on-demand article summarization and podcast generation pipeline
- **Let the Particles Have Babies! - PSO + GA + NN Mapping** Feb. 2024 - Jun. 2024
Tools: Python [Project]
◦ Combined Particle Swarm Optimization and Genetic Algorithms with neural networks for search-space reconstruction
- **Real-Time Multi-Filter App** Nov. 2022 - Dec. 2023
Tools: Python, NumPy [Project]
◦ Developed 20+ stackable image filters (emboss, oil paint, pencil sketch) in pure NumPy
◦ Optimized vectorized operations to maintain 60+ FPS on 1080p frames
- **Glomerular Detection & Disease Classification** Nov. 2022 - Dec. 2023
Tools: Python, PyTorch [Report Video]
◦ Accelerated segmentation pipeline from 364 to 29 days via entropy-based filtering
- **Taiwanese Language Learning App Based on LLMs** Sep. 2022 - Mar. 2024
Tools: Unity, Python, SpeechRecognition, TTS [Project]
◦ Integrated Unity-based AR with an LLM-driven agent for immersive language-learning scenarios
◦ Leveraged vision-language action commands to control interactive virtual agent
◦ Developed real-time speech recognition and synthesis pipelines for pronunciation feedback

EXPERIENCE

- **TAICA-Introduction to Artificial Intelligence** [🌐] Fall 2025
Teaching Assistant Tainan, Taiwan
 - Selected course under the TAICA Program with students from 50+ universities.
 - Assisted in courses and assignments design.
- **NCKU CSIE-Linear Algebra** [🌐] Spring 2024
Teaching Assistant Tainan, Taiwan
 - Assisted in courses and graded assignments.
- **NXP Semiconductors** [🌐] Feb. 2023 – Aug. 2023
Industry Intern Kaohsiung, Taiwan
 - Developed automated data preprocessing tools, algorithms, and logic validation scripts for large-scale data.
 - Enhanced existing pipelines to improve performance and maintainability.
- **Scratch Programming Instructor** Summer 2021 & 2022
Instructor Hsinchu, Taiwan
 - Taught introductory Scratch programming to elementary students, boosting communication and teaching skills.

HONORS AND AWARDS

- **Best Team Award** 2024
Critical Care Data Science Conference & Taiwan Datathon
 - Recognized for outstanding team collaboration and high-performance ICU data analysis with senior clinicians.

SKILLS

- **Programming Languages:** Python, C/C++, JavaScript, HTML, CSS
- **Data Science & Machine Learning:** NumPy, Pandas, PyTorch, TensorFlow, scikit-learn, Matplotlib
- **Developer Tools:** Docker, ROS2, Unity, React Native, Git, Bash
- **Specialized Areas:** Parameter-Efficient Fine-Tuning (LoRA), Adaptive Multi-Task Vision Models, Large Multi-Modal Model, App development